

Total Questions: 35

LEVEL – II

Set - B

Time: 1.5 hr.

Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

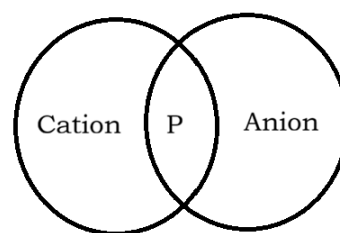
- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A swimmer underwater sees the Sun slightly shifted from its actual position. If refractive index of water increases, the shift will:
(A) Increase (B) Decrease
(C) Remain same (D) Reverse
2. Study the given the diagram which compares the properties of cation and anion.



Which of the following statements can be placed in P region?

- I. Formation leads to attainment of a stable electronic configuration.
- II. Formation depends on the ease of electron transfer.

III. They are formed by gain of electrons.

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II and III only

3. Ravi tries to wash an oily stain from a cloth using plain water but finds that the stain does not get removed. When soap is added, the stain disappears after some time.



What is the most appropriate explanation for this observation?

- (A) Soap reacts chemically with oil to form a new substance
 - (B) Soap forms micelles where hydrophobic ends trap oil and hydrophilic ends interact with water
 - (C) Water dissolves oil after soap increases its temperature
 - (D) Soap converts oil into a solid substance that can be removed easily
4. In a polluted city, the sky appears whitish instead of blue. The most appropriate explanation is:
- (A) Increase in wavelength of light
 - (B) Larger particles scatter all wavelengths nearly equally
 - (C) Absorption of blue light
 - (D) Decrease in intensity of sunlight

5. Which of the following correctly describes the "Double Circulation" in humans and explains why it is an evolutionary advantage over the "Single Circulation" found in fish?

- (A) Double circulation involves two hearts, one for oxygenated and one for deoxygenated blood, doubling the energy output.
- (B) It ensures that oxygenated and deoxygenated blood never mix, allowing for high-pressure delivery of oxygen to meet the high metabolic demands of warm-blooded animals.
- (C) It allows the blood to pass through the liver twice, ensuring that all toxins are removed before the blood reaches the brain.
- (D) It reduces the work of the lungs by allowing the skin to perform 50% of the gas exchange.

6. A touch-me-not plant (*Mimosa pudica*) folds its leaves almost instantly when touched. How does this "Nastic" movement differ fundamentally from the "Tropic" movement of a plant growing toward light?

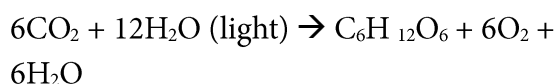
- (A) Nastic movements are growth-dependent, while tropic movements are independent of growth.
- (B) Nastic movements are directional, while tropic movements are non-directional.
- (C) Nastic movements are rapid and independent of growth, while tropic movements are slow and involve growth.

(D) Nastic movements involve the use of muscles, while tropic movements involve nerves.

7. The "Blood-Brain Barrier" is a highly selective semipermeable border of endothelial cells. What is its primary evolutionary purpose in control and coordination?

(A) To protect the brain from circulating toxins and pathogens while allowing vital nutrients like glucose to pass.
 (B) To prevent red blood cells from entering the brain and causing a stroke.
 (C) To act as a secondary skull to provide mechanical cushioning to the cerebrum.
 (D) To stop electrical impulses from leaving the brain and entering the spinal cord.

8. Consider the chemical equation for Photosynthesis. If the Oxygen gas released by a plant is found to contain the isotope ^{18}O , but the Carbon Dioxide supplied to the plant contains only the standard ^{16}O , what does this prove about the reaction?



(A) The oxygen gas released comes from the splitting of Water molecules (H_2O), not from CO_2 .
 (B) The plant is converting CO_2 directly into O_2 without using any water.
 (C) The glucose produced must also contain the ^{18}O isotope.
 (D) The reaction is unbalanced, and light energy is being converted into mass.

9. An object and its real image formed by a concave mirror are separated by 40 cm. The image is inverted and twice the size. Find focal length.

(A) -10.67 cm (B) 20.67 cm
 (C) 15.67 cm (D) -26.67 cm

10. Four statements about allotropes of carbon are given below. P. Diamond is the hardest natural substance and does not conduct electricity.

Q. Graphite is soft, slippery, and conducts electricity due to its layered structure.

R. Diamond and graphite are made of different elements and produce different oxides on burning.

S. Buckminsterfullerene is made up of only hexagonal rings of carbon atoms.

Select the correct combination of true (T) and false (F) statements.

	P	Q	R	S
(A)	T	T	F	F
(B)	T	F	T	F
(C)	F	T	F	T
(D)	T	T	T	F

11. A solenoid has 500 turns, length 0.25 m, current decreases uniformly from 4 A to 0 in 0.02 s. Find induced emf. (area = $1 \times 10^{-4} \text{ m}^2$)

(A) 0.025 V (B) 0.045 V
 (C) 0.085 V (D) 0.165 V

12. Which of the following sequences correctly represents the pathway of a sperm cell in the human male reproductive system, and what is the functional importance of the secretions added along the way?

(A) Testis → Epididymis → Vas deferens → Urethra; Secretions provide nutrition and a fluid medium for easier transport. wall.

(B) Testis → Vas deferens → Ureter → Urethra; Secretions help in digesting the egg's

(C) Scrotum → Kidney → Vas deferens → Penis; Secretions maintain the body temperature.

(D) Testis → Urethra → Vas deferens → Ureter; Secretions are used to kill bacteria in the female tract.

13. Consider a species of flowering plant where the pollen grains are heavy, sticky, and the flowers are brightly colored with a strong scent. If this plant is moved to an environment where its specific insect pollinator is extinct, but the wind is very strong, what is the most likely reproductive challenge it will face?

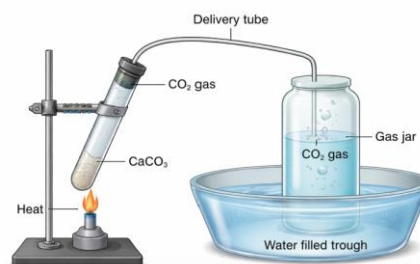
(A) It will easily switch to wind pollination because the sticky pollen will catch the wind better.

(B) The plant will immediately start producing smaller, lighter pollen grains to adapt to the wind.

(C) Its rate of cross-pollination will drop significantly because its structural adaptations are specific to biotic agents, not abiotic ones.

(D) The ovaries will stop producing ovules because they can sense the absence of insects in the environment.

14. A student heats 75 g of CaCO_3 and collects the carbon dioxide gas at STP.



What volume of CO_2 gas will be obtained at STP?

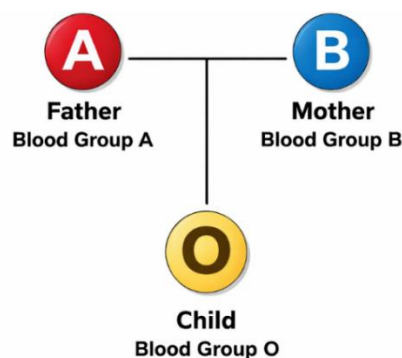
(A) 11.2 L

(B) 16.8 L

(C) 22.4 L

(D) 33.6 L

15. Study the following pedigree chart representation: A father has Blood Group 'A' and a mother has Blood Group 'B'. They have a child with Blood Group 'O'. Based on this diagrammatic data, what are the chances that their next child will have Blood Group 'AB', and what does this tell us about the alleles I^A and I^B ?



(A) 0% I^A and I^B are recessive to the 'O' allele.

(B) 100% I^A and I^B always merge to form a new intermediate blood type.

(C) 25% I^A and I^B are co-dominant, and both parents must be heterozygous ($I^A i$ and $I^B i$).

(D) 50% the 'O' group child proves that one parent must be ii .

Section - 2

16. A battery of emf V and internal resistance

$r = R$ is connected to an external resistance

R . Initial power across the external resistor is P . Now the external resistance is changed to $2R$.

Which of the following are correct?

- (A) Current becomes $2/3$ of initial
- (B) Power across external resistance becomes $8P/9$
- (C) Terminal voltage increases
- (D) Power supplied by battery decreases

17. Consider the following compound-type pairs.

- K. Methane — Saturated
- L. Ethene — Unsaturated
- M. Propane — Unsaturated
- N. Ethyne — Unsaturated
- O. Ethane — Saturated
- P. Propene — Saturated

Select the options that have correctly matched pairs.

- (A) L, N, O
- (B) L only
- (C) K
- (D) K, L, M

18. In his experiments, Gregor Mendel formulated the "Law of Independent Assortment." Which of the following conditions or observations support this law in a dihybrid cross ($RrYy \times RrYy$)?

- (A) The inheritance of the seed shape (Round/wrinkled) does not influence the inheritance of the seed color (Yellow/green).
- (B) The alleles for different characters are located on different pairs of homologous chromosomes.

(C) New combinations of traits (recombinants), such as Round-green or wrinkled-Yellow, appear in the F_2 generation.

(D) The phenotypic ratio of a dihybrid cross will always be 3:1 regardless of how many traits are studied simultaneously.

19. A long straight wire carries 15 A current.

A rectangular loop (current = 3 A, length = 0.2 m, width = 0.1 m) is placed such that its nearer side is at 5 cm and farther side at 15 cm.

Which of the following are correct? (APPROX)

- (A) Force on nearer side = $3.6 \times 10^{-5} \text{ N}$
- (B) Force on farther side = $1.2 \times 10^{-5} \text{ N}$
- (C) Force on farther side = $5.2 \times 10^{-5} \text{ N}$
- (D) Net force magnitude = $2.4 \times 10^{-5} \text{ N}$

20. Consider the following statements:

I. Phenolphthalein remains colourless in acidic solution.

II. Methyl orange turns red in basic solution.

III. Red litmus turns blue in sodium hydroxide solution.

IV. Turmeric indicator turns reddish-brown in soap solution.

Which of the following options correctly identifies the true and false statements?

- (A) II is false
- (B) III is true
- (C) I, II and III are only true
- (D) I, III and IV are only true

21. Which of the following are "Involuntary Actions" controlled primarily by the Hindbrain (Medulla and Pons)?

- (A) The peristaltic movement of food through the esophagus.
- (B) The sudden watering of the mouth (salivation) when smelling delicious food.
- (C) The precise movement of fingers while playing a complex piece on the piano.
- (D) The regulation of the breathing rate and blood pressure.

22. Consider the "Double Fertilization" event in a flowering plant (Angiosperm). Which of the following are the actual outcomes of this process?

- (A) The fusion of one male gamete with the egg cell results in a diploid zygote ($2n$).
- (B) The fusion of the second male gamete with two polar nuclei results in a triploid endosperm ($3n$).
- (C) The ovule gradually develops a tough coat and is converted into a seed.
- (D) The petals, sepals, and stamens grow larger to protect the developing fruit.

23. Which of the following substances are "Selectively Reabsorbed" from the glomerular filtrate back into the blood in a healthy human nephron?

- (A) Glucose
- (B) Urea
- (C) Amino acids
- (D) Water (in varying amounts depending on hydration)

Section - 3

COMPREHENSION-1

In a poultry farm, a breeder is investigating the inheritance of "Feather-Foot" (F), where feathers grow down the legs of chickens. This is a dominant trait.

The alternative is "Clean-Foot" (f), where the legs are smooth.

The breeder has two chickens:

- Hen A: A Feather-Foot chicken of unknown genotype.
- Rooster B: A Clean-Foot chicken.

To determine Hen A's genotype, the breeder performs a **Test Cross** by mating Hen A with Rooster B. They produce a total of 24 chicks. Out of these, 11 chicks have Clean-Foot and 13 chicks have Feather-Foot.

24. Based on the numerical data from the Test Cross (11:13), what is the most likely genotype of Hen A, and what biological law supports this?

- (A) FF (Homozygous Dominant); Law of Dominance.
- (B) Ff (Heterozygous); Law of Segregation.
- (C) ff (Homozygous Recessive); Law of Independent Assortment.
- (D) FfF (Triploid); Law of Co-dominance.

25. If the breeder selects two "Feather-Foot" chicks from the offspring (both necessarily heterozygous) and crosses them together, what is the mathematical probability that a chick from this new cross will be "Clean-Foot"?

- (A) 100%
- (B) 75%
- (C) 50%
- (D) 25%

26. In a different scenario, if "Feather-Foot" was a sex-linked dominant trait located on the Z-chromosome (recall that in birds, males are ZZ and females are ZW), what would be the result of crossing a Feather-Foot female ($Z^F W$) with a Clean-Foot male ($Z^f Z^f$)?

- (A) All male chicks will be Feather-Foot and all female chicks will be Clean-Foot.
- (B) All female chicks will be Feather-Foot and all male chicks will be Clean-Foot.
- (C) 100% of all chicks will be Feather-Foot regardless of sex.
- (D) All chicks will be Clean-Foot because the male is recessive.

COMPREHENSION-2

Corrosion is a process in which metals are oxidised by losing electrons. Some of the methods used for prevention of corrosion are painting and electroplating.

27. An iron nail is partially dipped in water and exposed to air. After a few days, rust is observed mainly near the water-air interface. What is the most appropriate reason?



- (A) The entire nail acts only as cathode
 - (B) Oxygen availability is higher near the interface
 - (C) Iron does not react in fully submerged conditions
 - (D) Water prevents oxidation of iron
28. In an experiment, an iron rod is connected to a magnesium strip and placed in a moist environment. After some time, magnesium corrodes instead of iron. What is the reason?

- (A) Magnesium acts as a cathode
- (B) Iron has higher reactivity than magnesium
- (C) Magnesium acts as a sacrificial anode
- (D) Iron forms a protective oxide layer faster

29. Consider the following statements.

Statement 1: Copper develops a green coating when exposed to moist air.

Statement 2: Copper reacts with carbon dioxide, oxygen and water to form basic copper carbonate.

- (A) Statement 1 and Statement 2 are correct, and Statement 2 is the correct explanation of Statement 1.
- (B) Statement 1 and Statement 2 are correct, but Statement 2 is not the correct explanation of Statement 1.
- (C) Both statements are false.
- (D) Statement 1 is true, but Statement 2 is false.
- (D) (i) and (iv) only

COMPREHENSION-3

A compound microscope is set so that the final image is formed at the **near point (25 cm)**. The objective has focal length **1 cm**, the eyepiece has focal length **5 cm**, and the separation between the lenses is **20 cm**. In correct adjustment, the intermediate image lies at a position such that the eyepiece forms the final image at 25 cm. During a precision experiment, the object is slightly displaced so that its distance from the objective becomes **1.2 cm**. The eyepiece is NOT adjusted.

30. The position of the intermediate image formed by the objective is:

- (A) 4 cm (B) 5 cm
(C) 6 cm (D) 7 cm

31. The object distance for the eyepiece (distance of intermediate image from eyepiece) is:

- (A) 14 cm (B) 15 cm
(C) 16 cm (D) 17 cm

32. If the object is further shifted by 0.02 cm towards the focal point, the change in image distance v of the objective is closest to:

- (A) 0.5 cm (B) 1 cm
(C) 2 cm (D) 4 cm

Section - 4

33. Match Column A with Column B

Column A	Column B
(A) Two resistors of equal resistance connected in parallel	(P) Power becomes four times
(B) A wire is folded in half (doubled over on itself)	(Q) Resistance becomes one-fourth
(C) Potential difference is doubled while resistance remains constant	(R) Equivalent resistance becomes three times
(D) Three identical resistors connected in series	(S) Equivalent resistance becomes half

- (A) A-S, B-Q, C-P, D-R
(B) A-S, B-Q, C-R, D-P

- (C) A-Q, B-P, C-P, D-S
(D) A-R, B-Q, C-P, D-S

34. Match Column A with Column B

Type of Reaction	Example
(A) Decomposition reaction	(P) Mixing baking soda solution with lemon juice
(B) Displacement reaction	(R) Burning of hydrogen in the presence of oxygen
(C) Combination reaction	(Q) Iron nail placed in copper sulphate solution
(D) Double displacement reaction	(S) Electrolysis of water

- (A) A - R, B - Q, C - P, D - S
(B) A - Q, B - P, C - S, D - R
(C) A - S, B - Q, C - R, D - P
(D) A - R, B - Q, C - P, D - S

35. Match Column A with Column B

Column I (Brain Region)	Column II (Specialized Role)
(A) Hypothalamus	(P) Precision of voluntary movements and balance
(B) Cerebrum	(Q) Regulation of hunger, thirst, and body temperature
(C) Medulla Oblongata	(R) Site of sensory perception, memory, and reasoning
(D) Cerebellum	(S) Control of blood pressure, salivation, and vomiting

- (A) a-Q, b-R, c-S, d-P
(B) a-R, b-Q, c-P, d-S
(C) a-Q, b-P, c-S, d-R
(D) a-S, b-R, c-Q, d-P